PRACTICE SHEET

1. In a triangle ABC, a = 2b and $\angle A = 3\angle B$. Which one of the following is correct?
(a) The triangle is isosceles
(b) The triangle is equilateral
(c) The triangle is right – angled
(d) Such triangle does not exist

2. The median AD of a triangle ABC is bisected at F, and BF is produced to meet the side AC in P. If AP = λAC, then what is the value of λ?
(a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) $\frac{2}{3}$ (d) $\frac{1}{3}$

3. If the perimeter of a triangle ABC is 30cm, then what is the value of $a\cos^2(C/2) + c\cos^2(A/2)$?
(a) 15cm (b) 10cm
(c) $\frac{15}{2}$ cm (d) 13cm

4. In ΔABC, if $\angle A : \angle B : \angle C = 1 : 2 : 3$, then what is BC : CA: AB?
(a) 1 : 2 : 3 (b) $1 : \sqrt{3} : 2$
(c) $2 : \sqrt{3} : 1$ (d) $\sqrt{3} : 1 : 2$

5. The angles A, B, C of a triangle are in the ratio 2 : 5 : 5. What is the value of tan B tan C?
(a) $4+\sqrt{3}$ (b) $4+2\sqrt{3}$
(c) $7+4\sqrt{3}$ (d) $3+3\sqrt{3}$

6. In a triangle ABC, if a = 2b and A = 3B then which one of the following is correct?
(a) The triangle is obtuse – angled
(b)The triangle is acute –angled but not right –angled
(c) The triangle is right – angled
(d) The triangle is isosceles but not obtuse – angled

7. If median of the ΔABC through A is perpendicular to BC, then which one of the following is correct?
(a) tan A + tan B = 0
(b) tan B – tan C = 0
(c) tan C + 2 tan A = 0
(d) tan B + tan C = 0

8. In a triangle ABC, b = $\sqrt{3}$ cm, c = 1 cm, $\angle A = 30°$, what is the value of a?
(a) $\sqrt{2}$ (b) 2 cm
(c) 1 cm (d) $\frac{1}{2}$cm

9. ABC is a triangle in which AB = 6m, BC = 8 cm and CA = 10 cm. What is the value of cot (A/4)?
(a) $\sqrt{5}-2$ (b) $\sqrt{5}+2$
(c) $\sqrt{3}-1$ (d) $\sqrt{3}+1$

10. If the sides of a triangle are 6cm, 10cm and 14cm, then what is the largest angle included by the sides?
(a) 90° (b) 120°
(c) 135° (d) 150°

11. In a triangle ABC, AC = 5 cm, AB = 7 cm, BC = √3 . What is the measure of angle A ?
(a) 45^0 (b) 60^0
(c) 900 (d) 300

12. ABC is a triangle in which BC = 10 cm, CA = 6 cm and AB = 8 cm, which one of the following is a correct?
(a) ABC is an acute angled triangle
(b) ABC is an obtuse angled triangle
(c) ABC is an right angled triangle
(d) None of the above

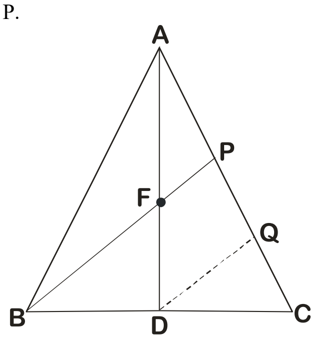
ANSWER KEY

1.	c	**2.**	d	**3.**	a	**4.**	b	**5.**	c	**6.**	c	**7.**	b	**8.**	b	**9.**	b	**10.**	b
11.	b	**12.**	c																

Solutions

Sol. 1. (c)
From properties of triangle we know that
$$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}$$
Given that a = 2b and A = 3B.
$$\text{We get}\,\frac{2b}{\sin 3B}=\frac{b}{\sin B}.$$
$$\Rightarrow \frac{\sin 3B}{\sin B}=2 \Rightarrow \frac{3\sin B-4\sin^3 B}{\sin B}=2$$
$$\Rightarrow 3-4\sin^2 B=2$$
$$\Rightarrow \sin^2 B=\left(\frac{1}{2}\right)^2 \Rightarrow \sin B=\frac{1}{2}=\sin\frac{\pi}{6}$$
$$\Rightarrow B=\frac{\pi}{6}\,\text{since}\,A=3B$$
$$\text{So,}\,\angle A=\frac{\pi}{2}$$
Thus, triangle is right angled triangle.

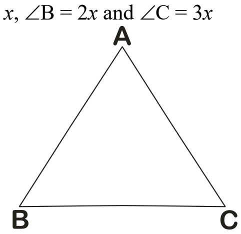
Sol. 2. (d)
Given, AD is the median of Δ ABC. F is mid – point of AD and BF is produced to meet the side AC in P.



Draw DQ || FP
In ΔADQ, F is mid – point of AD and FP || DQ.
∴ P is mid – point of AQ (converse of mid – point theorem)
⇒AP = PQ
In ΔBCP, D is mid – point of BC and DQ|| BP.
∴Q is midpoint of PC
⇒PQ = QC
From (1), (2) we get, AP = PQ = QC
From figure, AP + PQ + QC = AC
⇒AP + AP + AP = AC
⇒ 3× AP = AC
$$\Rightarrow AP=\frac{1}{3}\times AC \quad \therefore \lambda=\frac{1}{3}$$

Sol. 3. (a)
We know from properties of triangle that
$$\cos\frac{A}{2}=\sqrt{\frac{s(s-a)}{bc}} \Rightarrow \cos^2\frac{A}{2}=\frac{s(s-a)}{bc}$$
$$\text{and}\cos\frac{C}{2}=\sqrt{\frac{s(s-c)}{ab}} \Rightarrow \cos^2\frac{C}{2}=\frac{s(s-c)}{ab}$$
$$\text{So,}\,a\cos^2\frac{C}{2}+c\cos^2\frac{A}{2}$$
$$=a\left(\frac{s(s-c)}{ab}\right)+c\left(\frac{s(s-a)}{bc}\right)=\frac{s(s-c+s-a)}{b}$$
$$=\frac{s(2s-a-c)}{b}=\frac{s(a+b+c-a-c)}{b}=\frac{s.b}{b}=s$$
$$=\frac{30}{2}=15\text{cm} \qquad [\text{given that}\,2s=30]$$

Sol. 4. (b)
Ratio of angles is given by ∠A : ∠B : ∠C = 1 : 2 : 3
Let ∠A = x, ∠B = $2x$ and ∠C = $3x$

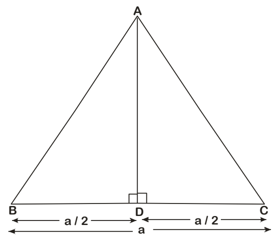


We know that in a triangle
∠A + ∠B + ∠C = 180°
⇒ $x + 2x + 3x$ = 180°
$$\Rightarrow x=\frac{180°}{6}=30°$$
So, ∠A = 30°, ∠B = 60° and ∠C = 90°
From sin law
$$\frac{BC}{\sin A}=\frac{CA}{\sin B}=\frac{AB}{\sin C}=K$$
$$\frac{BC}{\sin 30°}=\frac{CA}{\sin 60°}=\frac{AB}{\sin 90°}=K$$
BC = K sin 30°
CA = K sin 60°
AB = K sin 90°
BC : CA : AB
= sin 30° : sin 60° : sin 90°
$$=\frac{1}{2}:\frac{\sqrt{3}}{2}:1=1:\sqrt{3}:2$$

Sol. 5. (c)
Let the angles A, B and C of a triangle are $2x$, $5x$ and $5x$, respectively
So, $2x + 5x + 5x$ = 180°
$$\Rightarrow x=\frac{180°}{12}=15°$$
Angles are 30°, 75°, 75°
∠B = 75° and ∠C = 75°
∴ tan B and C = $(\tan 75°)^2 = (\tan(45°+30°))^2$
$$=\left(\frac{\tan 45°+\tan 30°}{1-\tan 45°\tan 30°}\right)=\left(\frac{1+\frac{1}{\sqrt{3}}}{1-\frac{1}{\sqrt{3}}}\right)=\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)^2$$
$$=\left(\frac{\sqrt{3}+1}{3-1}\right)^2=\frac{1}{4}\left[3+1+2\sqrt{3}\right]^2$$
$=1/4[4+2\sqrt{3}]^2 = \frac{1}{4}[16+12+16\sqrt{3}]$
$=1/4[28+16\sqrt{3}]=7+4\sqrt{3}$

Sol. 6. (c)
We know from the sine law that
$$\frac{a}{\sin A}=\frac{b}{\sin B}$$
$$\Rightarrow \frac{2b}{\sin 3B}=\frac{b}{\sin B}$$
⇒ 2 sin B = sin 3 B
⇒ 2 sin B = 3 sin B – 4 $\sin^3$ B
⇒ sin B – 4 $\sin^3$B = 0
⇒ sin B (1 – 4 $\sin^2$B) = 0
⇒ sin B = 0 or 1 – 4 $\sin^2$B = 0 ⇒ B = 0 or B = 30°
⇒ B = 30° and A = 3 × 30° = 90°
⇒ B = 0 is not possible so, B = 30° and A = 3 × 30° = 90°
⇒ The triangle is right angled triangle.

Sol. 7. (b)
In the given ΔABC
Let BC = a



$$\therefore BD=CD=\frac{a}{2}$$
In ΔADB,
$$\tan B=\frac{AD}{BD}=\frac{AD}{a/2}$$
$$\Rightarrow \tan B=\frac{2AD}{a} \qquad \ldots(1)$$
In ΔADC
$$\tan C=\frac{AD}{CD}=\frac{AD}{a/2}$$
$$\tan C=\frac{2AD}{a} \qquad \ldots.(2)$$
From eqs.(1) and(2), we get tan B=tan C
⇒ tan B – tan C = 0

Sol. 8. (c)
As given : In a triangle ABC, AC = b = $\sqrt{3}$ cm, AB = c = 1m and ∠A = 30°
From cosine formulae

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{(\sqrt{3})^2 + 1^2 - a^2}{2\sqrt{3}.1}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{3 + 11 - a^2}{2\sqrt{3}} \Rightarrow 3 = 4 - a^2$$

$\Rightarrow a^2 = 4 - 3 = 1 \Rightarrow a = 1\text{cm}$

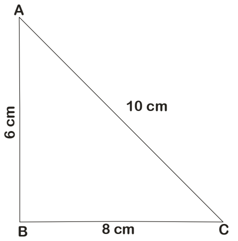
Sol. 9. **(b)**
Here, AB = 6 cm, BC = 8cm and CA = 10cm

$$S = \frac{a + b + c}{2} = \frac{24}{2} = 12$$

$$\therefore \tan\frac{A}{2} = \sqrt{\frac{(12-10)(12-6)}{12(12-8)}}$$

$$\left(\because \tan\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}\right)$$

$$= \sqrt{\frac{1}{4}} = \frac{1}{2}$$



$\therefore \cot A/2 = 2$

$$\text{Now, } \cot\left(\frac{A}{4} + \frac{A}{4}\right) = \frac{\cot^2\frac{A}{4} - 1}{2\cot\frac{A}{4}}$$

$$\cot\left(\frac{A}{2}\right) = \frac{\cot^2\frac{A}{4} - 1}{2\cot\frac{A}{4}}$$

$$\text{Let } \cot\left(\frac{A}{4}\right) = x$$

$$\therefore 2 = \frac{x^2 - 1}{2x}$$

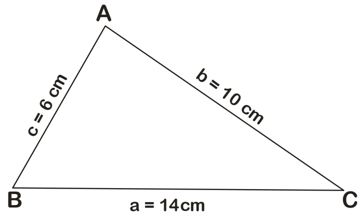
$\Rightarrow x^2 - 4x - 1 = 0$

$$\Rightarrow x = \frac{4 \pm \sqrt{16 + 4}}{2}$$

$$\Rightarrow x = \frac{4 \pm 2\sqrt{5}}{2} = 2 \pm \sqrt{5}$$

$$\text{So, } \cot\left(\frac{A}{4}\right) = \sqrt{5} + 2 \text{ or } 2 - \sqrt{5}$$

Sol. 10. **(b)**
We know that largest side has greatest angle opposite it,
$\therefore$ a = 14cm, b = 10cm and c = 6cm



$$\therefore \cos A = \frac{c^2 + b^2 - a^2}{2bc}$$

$$= \frac{36 + 100 - 96}{2 \times 6 \times 10} = -\frac{1}{2} = \cos 12°$$

$\Rightarrow \angle A = 120°$

Sol. 11. **(b)**

$$CosA = \frac{b^2 + c^2 - a^2}{2bc} = \frac{25 + 49 - 39}{2 \times 5 \times 7}$$

CosA = 1/2
A = 60^0

Sol. 12. **(c)**
$6^2 + 8^2 = 10^2$
so triangle is right angles

NDA PYQ

Direction (For next four):
ABC is a triangle right angled at B. The hypotenuse (AC) is four times the perpendicular (BD) drawn to it from the opposite vertex and AD < DC.

1. What is one of the acute angles of the triangle?
(a) 15^0 (b) 30^0
(c) 45^0 (d) none
[NDA (I) - 2011]

2. What is $\angle ABD$?
(a) 15^0 (b) 30^0
(c) 45^0 (d) none
[NDA (I) - 2011]

3. What is AD : DC equal to?
(a) $\left(7-2\sqrt{3}\right):1$ (b) $\left(7-4\sqrt{3}\right):1$
(c) 1:2 (d) none of these
[NDA (I) - 2011]

4. What is tan (A − C) equal to?
(a) 0 (b) 1
(c) 2 (d) none
[NDA (I) - 2011]

5. In a triangle ABC, if c = 2, A = 120^0 and a = $\sqrt{6}$, then what is angle C equal to
(a) 75^0 (b) 60^0
(c) 45^0 (d) 30^0
[NDA (I) - 2011]

6. If the angles of a triangle ABC are in AP and b : c = $\sqrt{3}$: $\sqrt{2}$, then what is the measure of angle A?
(a) 30^0 (b) 45^0
(c) 60^0 (d) 75^0
[NDA (I) - 2011]

7. The sides a, b, c of a ΔABC are in AP and 'a' is the smallest side. What is cos A equal to?
(a) $\frac{3c-4b}{2c}$ (b) $\frac{3c-4b}{2b}$
(c) $\frac{4c-3b}{2c}$ (d) $\frac{3b-4c}{2c}$
[NDA (II) - 2011]

8. In a ΔABC, a = 8, b = 10 and c = 12. What is $\angle C$ equal?
(a) A/2 (b) 2A
(c) 3A (d) 3A /2
[NDA (II) - 2011]

9. If sides of a triangle are in the ratio 2: $\sqrt{6}$: 1+ $\sqrt{3}$, then what is the smallest angle of the triangle?
(a) 75^0 (b) 60^0
(c) 45^0 (d) 30^0
[NDA (II) - 2011]

10. If the sides of a triangle are 6cm, 10cm and 14cm. Then the triangle is obtuse angled with the obtuse angle equal to
(a) 150^0 (b) 135^0
(c) 120^0 (d) 105^0
[NDA (II) - 2011]

11. In a ΔABC, if the angles A, B, C are in AP, then which one of the following is correct?
(a) c = a + b (b) $c^2 = a^2 + b^2 - ab$
(c) $a^2 = b^2 + c^2 - bc$ (d) $b^2 = a^2 + c^2 - ac$
[NDA (I) - 2012]

12. In any ΔABC, a = 18, b = 24 and c = 30, then what is Sin C equal?
(a) 1/4 (b) 1/3
(c) 1/2 (d) 1
[NDA (I) - 2013]

13. If the angles of a triangle are 30^0, 45^0 and the included side is $\sqrt{3}$ + 1, then what is the area of triangle?
(a) $\left(\frac{\sqrt{3}+1}{2}\right)cm^2$ (b) $\left(2\sqrt{3}+1\right)cm^2$
(c) $\left(\frac{\sqrt{3}+1}{3}\right)cm^2$ (d) $\left(\frac{\sqrt{3}-1}{2}\right)cm^2$
[NDA (II) - 2013]

14. Consider the following statements:
I.There exists no ΔABC for which Sin A + Sin B = Sin C
II.If the angles of a triangle are in the ratio 1 : 2 : 3, then its sides will be in the ratio 1 : $\sqrt{3}$: 2
Which of the above statement(s) is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
[NDA (I) - 2014]

15. In a ΔABC, if c = 2, A = 45^0 and a = $2\sqrt{2}$, then what is C equal to?
(a) 30^0 (b) 15^0
(c) 45^0 (d) none
[NDA (II) - 2014]

16. In a triangle ABC, sinA – cosB = cosC, then what is B equal to?
(a) π (b) π/3
(c) π/2 (d) π/4
[NDA-2014(2)]

17. In a ΔABC , a = (1+ $\sqrt{3}$), b = 2 cm, and ∠C = 60°, then the other two angles are
(a) 45° and 75° (b) 30° and 90°
(c) 105° and 15° (d)100°and 20°
[NDA (I) - 2015]

Direction (for next two): Consider a ΔABC satisfying
$$2a\sin^2\left(\frac{C}{2}\right)+2c\sin^2\left(\frac{A}{2}\right)=2a+2c-3b$$

18. The sides of the triangle are in
(a) GP (b) AP
(c) HP (d) None
[NDA (II) - 2015]

19. Sin A, Sin B, Sin C are in
(a) GP (b) AP
(c) HP (d) None
[NDA (II) - 2015]

20. If a, b, c are the sides of a triangle ABC, then $a^{\frac{1}{p}}+b^{\frac{1}{p}}-c^{\frac{1}{p}}$ where p>1, is :
(a) always negative
(b) always positive
(c) always zero
(d) positive if 1 < p < 2 and negative if p >2
[NDA-2015(2)]

21. Consider the following statements:
1.If ABC is an equilateral triangle, then 3tan (A+B) tan C=1
2.If ABC is a triangle in which A=78°, B=66°, then

$\tan\left(\frac{A}{2}+C\right) < \tan A$

3.If ABC is any triangle, then $\tan\left(\frac{A+B}{2}\right)\sin\left(\frac{C}{2}\right) < \cos\left(\frac{C}{2}\right)$

Which of the above statements is/are correct?

(a) 1 only (b) 2 only
(c) 1 and 2 (d) 2 and 3

[NDA-2016(1)]

Direction (for next two): Consider a ΔABC in which $\cos A + \cos B + \cos C = \sqrt{3}\,Sin\frac{\pi}{3}$

22. What is the value of $Sin\left(\frac{A}{2}\right)Sin\left(\frac{B}{2}\right)Sin\left(\frac{C}{2}\right)$

(a) 1/2 (b) 1/4
(c) 1/8 (d) 1/8

[NDA (I) - 2016]

23. What is the value of $Cos\left(\frac{A+B}{2}\right)Cos\left(\frac{B+C}{2}\right)Cos\left(\frac{C+A}{2}\right)$

(a) 1/4 (b) 1/2
(c) 1/16 (d) None

[NDA (I) - 2016]

24. Consider the following for triangle ABC

I. $Sin\left(\frac{B+C}{2}\right) = Cos\left(\frac{A}{2}\right)$

II. $\tan\left(\frac{B+C}{2}\right) = Cot\left(\frac{A}{2}\right)$

III. Sin(B + C) = Cos A

IV. tan(B + C) = − Cot A

Which of the above are correct ?

(a) I and III (b) I and II
(c) I and IV (d) II and III

[NDA (I) - 2017]

25. In a ΔABC, a − 2b + c = 0, then the value of $\cot\left(\frac{A}{2}\right)\cot\left(\frac{C}{2}\right)$

(a) 9/2 (b) 3
(c) 3/2 (d) 1

[NDA (II) - 2017]

26. In a ΔABC, If $\frac{Sin^2A+Sin^2B+Sin^2C}{Cos^2A+Cos^2B+Cos^2C} = 2$ then the triangle is

(a) right angled (b) equilateral
(c) isosceles (d) obtuse angled

[NDA (II) - 2017]

27. If x, x–y and x+y are the angles of a triangle (not an equilateral triangle) such that tan (x–y), tan x and tan (x+y) are in GP, then what is x equal to?

(a) π/4 (b) π/3
(c) π/6 (d) π/2

[NDA-2018(1)]

28. In a ΔABC, If a = 2, b = 3, and sinA = 2/3 then what is value of angle B

(a) 45° (b) 90°
(c) 60° (d) 30°

[NDA (I) - 2018]

29. ABC is a triangle inscribed in a circle with centre O, Let α = ∠BAC, where 45°< α <90°, let β = ∠BOC , which one of the following is correct?

(a) $\cos\beta = \frac{1-\tan^2\alpha}{1+\tan^2\alpha}$ (b) $\cos\beta = \frac{1+\tan^2\alpha}{1-\tan^2\alpha}$
(c) $\cos\beta = \frac{2\tan\alpha}{1+\tan^2\alpha}$ (d) $Sin\beta = 2Sin^2\alpha$

[NDA (I) - 2018]

30. If (A + B + C) = 180^0 , then what is Sin2A − Sin2B −Sin2C =

(a) −4SinA.SinB.SinC (b) −4CosA.SinB.CosC
(c) −4CosA.CosB.SinC (d) −4SinA.CosB.CosC

[NDA (II) - 2018]

31. If the angles of a triangle ABC are in the ratio 1:2:3, then the corresponding sides are in the ratio?

(a) 1 : 2 : 3 (b) 3 : 2 : 1
(c) 1 : $\sqrt{3}$: 2 (d) 1 : $\sqrt{3}$: $\sqrt{2}$

[NDA (I) - 2019]

32. If angle C of a triangle ABC is a right angle, then what is tanA + tanB equal to?

(a) $\frac{a^2-b^2}{ab}$ (b) $\frac{a^2}{bc}$
(c) $\frac{b^2}{cà}$ (d) $\frac{c^2}{ab}$

[NDA-2019(2)]

33. If the angles of a triangle ABC are in AP and b : c = $\sqrt{3}$: $\sqrt{2}$, then what is the measure of angle A?

(a) 30^0 (b) 45^0
(c) 60^0 (d) 75^0

[NDA (II) - 2019]

34. Consider the following statements :

1.If in a triangle ABC, A = 2B and b = c , then it must be an obtuse angled triangle.

2.There exists no triangle ABC with A = 40°, B = 65° and $\frac{a}{c} = \sin 40° \cos ec 15°$

Select the correct answer using the code given below

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA 2020]

35. Consider the following statements :

1. If ABC is a right angled triangle, right angled at A and if SinB =1/3, then cosecC= 3

2. If b cosB = c cosC and if the triangle ABC is not an right angled then ABC must be isossceles

Select the correct answer using the code given below

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA 2020]

Direction (For next two):
Read the following information and answer the given items
ABCD is a trapezium such that AB and CD are parallel and BC is perpendicular to them, Let ∠ABD= α, ∠ADB= θ, BC = p and CD = q

36. Consider the following:

1. AD sin θ = AB sin α
2. BD sin θ = AB sin(θ + α)

Select the correct answer using the code given below

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA 2020]

37. What AB equal to?

(a) $\frac{(p^2+q^2)\sin\theta}{p\cos\theta+q\sin\theta}$ (b) $\frac{(p^2-q^2)\sin\theta}{p\cos\theta+q\sin\theta}$

(c) $\frac{(p^2+q^2)\sin\theta}{q\cos\theta+p\sin\theta}$ (d) $\frac{(p^2-q^2)\sin\theta}{q\cos\theta+p\sin\theta}$

[NDA 2020]

38. A chord subtends an angle 120° at the centre of a unit circle. What is the length of the chord?

(a) $\sqrt{2}-1$ units (b) $\sqrt{3}-1$ units

(c) $\sqrt{2}$ units (d) $\sqrt{3}$ units

[NDA 2021 (I)]

39. The sides of a triangle are m, n and $\sqrt{m^2+n^2+mn}$. What is the sum of the acute angles of the triangle?

(a) 45° (b) 60°

(c) 75° (d) 90°

[NDA 2021 (I)]

40. What is the area of the triangle ABC with sides a = 10 cm, c = 4 cm and angle B = 30°?

(a)16 cm^2 (b) 12 cm^2

(c) 10cm^2 (d) 8cm^2

[NDA 2021 (I)]

41. Let ABC be a triangle. If cos2A + cos2B + cos 2C = – 1. then which one of the following is correct?

(a) sinA sinB sinC = 0 (b) sinA sinB cosC = 0

(c) cosA sinB sinC = 0 (d) cosA cosB cosC = 0

[NDA (II) - 2021]

42. In a triangle ABC, sinA – cosB – cos C = 0. What is single B equal to?

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

[NDA 2022 (I)]

43. Let a, b, c be the lengths of sides BC, CA, AB respectively of a triangle ABC. If p is the perimeter and q is the area of the triangle, then what is $p(p-2a)\tan\left(\frac{A}{2}\right)$ equal to?

(a) q (b) 2q

(c) 3q (d) 4q

[NDA 2022 (I)]

Consider the following for the next three (03) items that follow:

ABC is a triangular plot with AB = 16m, BC = 10m and CA =10m. A lamp post is situated at the middle point of the side AB. The lamp post subtends an angle 45° at the vertex B.

44. What is the height of the lamp post?

(a) 6m (b) 7m

(c) 8m (d) 9m

[NDA 2022 (II)]

45. What is $\frac{AB}{\sin C}$ equal to?

(a) 17m (b) $\frac{50}{3}$ m

(c) $\frac{40}{3}$ m (d) 16m

[NDA 2022 (II)]

46. What is cosA + cosB + cosC equal to?

(a) 1 (b) $\frac{41}{25}$

(c) $\frac{37}{25}$ (d) $\frac{33}{25}$

[NDA 2022 (II)]

47. In a triangle ABC, a = 4, b = 3, c = 2. What is cos3C equal to?

(a) $\frac{7}{128}$ (b) $\frac{11}{128}$

(c) $\frac{7}{64}$ (d) $\frac{11}{64}$

[NDA 2022 (II)]

Consider the following for the next two (02) items that follow:

In a triangle PQR, P is the largest angle and cosP $=\frac{1}{3}$.

Further the in-circle of the triangle touches the sides PQ, QR and RP at N, L and M respectively such that the length PN, QL and RM are n, n + 2, n + 4 respectively where n is an integer.

48. What is the value of n?

(a) 4 (b) 6

(c) 8 (d) 10

[NDA – 2023 (1)]

49. What is the length of the smallest side?

(a) 12 (b) 14

(c) 16 (d) 18

[NDA – 2023 (1)]

Consider the following for the next two (02) items that follow:

The perimeter of a triangle ABC is 6 times the AM of sine of angles of the triangle. Further BC = √3 and CA = 1.

50. What is the perimeter of the triangle?

(a) √3+1 (b) √3+2

(c) √3+3 (d) 2√3+1

[NDA – 2023 (1)]

51. Consider the following statements:

1.ABC is right angled triangle

2.The angles of the triangle are in AP

Which of the statements given above is/are correct?

(a)1 only (b) 2 only

(c) both 1 and 2 (d) neither 1 nor 2

[NDA – 2023 (1)]

Consider the following for the next two (02) items that follow:

In the triangle ABC, $a^2 + b^2 + c^2 = ac + \sqrt{3}\, bc$

52. What is the nature of the triangle?

(a) equilateral

(b) isosceles

(c) right angled triangle

(d) scalene but not right angles

[NDA – 2023 (1)]

53. If c = 8, what is the area of the triangle?

(a) 4√3 (b) 6√3

(c) 8√3 (d) 12√3

[NDA – 2023 (1)]

Consider the following for the next (02) items

The angles A, B and C of a triangle ABC are in the ratio 3:5:4.

54. What is the value of a + b + √2c equal to?

(a) 3a (b) 2b

(c) 3b (d) 2c

[NDA-2023 (2)]

55. What is the ratio of $a^2 : b^2 : c^2$?

(a) 2 : 2 + √3 : 3 (b) 2 : 2 – √3 : 2

(c) $2 : 2 + \sqrt{3} : 2$ (d) $2 : 2 - \sqrt{3} : 3$

[NDA-2023 (2)]

56. In a triangle ABC, AB = 16 cm, BC = 63 cm and AC = 65 cm. What is the value of cos 2A + cos 2B + cos 2C?

(a) –1 (b) 0

(c) 1 (d) $\frac{76}{65}$

[NDA-2024 (1)]

57. Consider the following statements

1. In a triangle ABC, if cotA·cotB·cotC > 0, then the triangle is an acute angled triangle

2. In a triangle ABC, if tanA·tanB·tanC > 0, then the triangle is an obtuse angled triangle

Which of the statements given above are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA-2024 (1)]

58. ABC is a triangle such that angle C = 60°, then what is $\frac{\cos A + \cos B}{\cos\left(\frac{A-B}{2}\right)}$ equal to?

(a) 2 (b) $\sqrt{2}$

(c) 1 (d) $\frac{1}{\sqrt{2}}$

[NDA-2024 (1)]

59. In a triangle ABC $\frac{a}{\cos A} = \frac{b}{\cos B} = \frac{c}{\cos C}$ what is the area of the triangle if a = 6 cm ?

(a) $9\sqrt{3}$ square cm (b) 12 square cm

(c) $18\sqrt{3}$ square cm (d) 24 square cm

[NDA-2024 (2)]

60. The roots of the equation $7x^2 - 6x + 1 = 0$ are tanα and tanβ, where 2α and 2β are the angles of a triangle. Which one of the following is correct ?

(a) The triangle is equilateral

(b) The triangle is isosceles but not right-angled

(c) The triangle is right-angled

(d) The triangle is right-angled isosceles

[NDA-2024 (2)]

61. In a triangle ABC, ∠A = 75° and ∠B = 45°. What is 2a – b equal to ?

(a) c (b) $\sqrt{2}c$

(c) 2c (d) $2\sqrt{2}c$

[NDA-2024 (2)]

62. In a triangle ABC tanA + tanB + tanC = k What is the value of cotAcotBcotC ?

(a) 0.5k (b) 1/k

(c) 3/k (d) $1/k^3$

[NDA-2024 (2)]

Consider the following for the three (03) items that follow:

The sides of a triangle are AB = 3 cm, BC = 5 cm, CA = 7 cm

63. Consider the following statements:

I. The triangle is obtuse-angled triangle.

II. The sum of acute angles of the triangle is also acute.

Which of the statements given above is/are correct?

(a) I only (b) II only

(c) Both I and II (d) neither I nor II

[NDA-2025 (1)]

64. What is ∠B equal to?

(a) 60° (b) 105°

(c) 120° (d) 150°

[NDA-2025 (1)]

65. What is the area of the triangle?

(a) $15\sqrt{3}$ /4 sq. cm (b) $15\sqrt{3}$ /2 sq. cm

(c) $15\sqrt{3}$ sq. cm (d) $30\sqrt{3}$ sq. cm

[NDA-2025 (1)]

Consider the following for the two (02) items that follow:

In a triangle ABC, two sides BC and CA are in the ratio 2:1 and their opposite corresponding angles are in the ratio 3:1.

66. One of the angles of the triangle is

(a) 15° (b) 30°

(c) 45° (d) 75°

[NDA-2025 (1)]

67. Consider the following statements

I. The triangle is right-angled.

II. One of the sides of the triangle is 3 times the other.

III. The angles A,C and B of the triangle are in A.P.

Which of the statements given above is/are correct?

(a) I only (b) II and III only

(c) I and III only (d) I, II and III

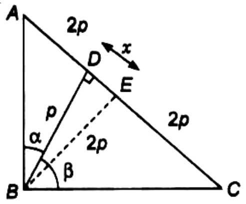
[NDA-2025 (1)]

ANSWER KEY

1.	a	**2.**	a	**3.**	b	**4.**	d	**5.**	c	**6.**	d	**7.**	c	**8.**	b	**9.**	c	**10.**	c
11.	d	**12.**	d	**13.**	a	**14.**	c	**15.**	a	**16.**	c	**17.**	a	**18.**	b	**19.**	b	**20.**	a
21.	b	**22**	c	**23.**	d	**24.**	b	**25.**	b	**26.**	a	**27.**	b	**28.**	b	**29.**	a	**30.**	d
31.	c	**32.**	d	**33.**	d	**34.**	b	**35.**	d	**36.**	c	**37.**	a	**38.**	d	**39.**	b	**40.**	c
41.	d	**42.**	d	**43.**	d	**44.**	c	**45.**	b	**46.**	d	**47.**	a	**48.**	c	**49.**	d	**50.**	c
51.	c	**52.**	c	**53.**	c	**54.**	c	**55.**	a	**56.**	a	**57.**	a	**58.**	c	**59.**	a	**60.**	c
61.	b	**62.**	b	**63.**	c	**64.**	c	**65.**	a	**66.**	b	**67.**	c							

Solutions

Sol.1. (a)
Let BD = p and DE = x



AC = 4p
Let E be a mid point of line AC.
Then AE = EC = BE = 2p
inΔBDE,$(BE)^2 = (BD)^2 + (ED)^2$
$(2p)^2 = p^2 + x^2$
x = $\sqrt{3}p$
Now AD = 2p − x = 2p − $\sqrt{3}p$
DC = 2p + x = 2p + $\sqrt{3}p$
In ΔBAD, $\tan A = \dfrac{BD}{AD} = \dfrac{p}{2p - p\sqrt{3}}$

$$\tan A = \frac{1}{2-\sqrt{3}} = 2+\sqrt{3}$$

A = 75^0 than C = 15^0
than α = 15^0 and β = 75^0

Sol.2.(a)
$\angle ABD = 15^0$

Sol.3.(b)
AD : DC = $\left(7-4\sqrt{3}\right):1$

Sol.4.(d)
tan(A − C) = tan(75^0−15^0) = tan60^0 =√3

Sol.5.(c)
by sine formula

$$\frac{SinA}{a} = \frac{SinC}{c} \Rightarrow \frac{Sin120^0}{\sqrt{6}} = \frac{SinC}{2}$$

by solving this C = 45^0

Sol.6.()
Angles A, B, C are in AP then angle B = 60°

$$\frac{\text{SinB}}{\text{b}} = \frac{\text{SinC}}{\text{c}}$$

$$\frac{\text{Sin}\,60}{\sqrt{3}} = \frac{\text{SinC}}{\sqrt{2}}$$

By solving it C = 45°than A= 75°

Sol.7.(c)
a, b, c are in AP
2b = a + c

$$CosA = \frac{b^2+c^2-(2b-c)^2}{2bc} = \frac{b^2+c^2-4b^2-c^2+4bc}{2bc}$$

$$CosA = \frac{4bc-3b^2}{2bc} = \frac{4c-3b}{2c}$$

Sol.8.(b)
a = 8 cm, b = 10 cm , c = 12 cm

$$CosC = \frac{a^2+b^2-c^2}{2ab}$$

$$CosC = \frac{8^2+10^2-12^2}{2.8.10} = \frac{1}{8}$$

now

$$CosA = \frac{b^2+c^2-a^2}{2bc}$$

$$CosA = \frac{10^2+12^2-8^2}{2.10.12} = \frac{3}{4}$$

Cos2A = $2\cos^2 A - 1 = 2(3/4)^2 - 1 = 1/8$
so C = 2A

Sol.9.(c)
Given $a:b:c = 2:\sqrt{6}:(\sqrt{3}+1)$
a is minimum so angle A will be smallest

$$\text{CosA} = \frac{b^2+c^2-a^2}{2bc}$$

$$\text{CosA} = \frac{6+(\sqrt{3}+1)^2-4}{2(\sqrt{6})(\sqrt{3}+1)}$$

$$\text{CosA} = \frac{6+4+2\sqrt{3}-4}{2(\sqrt{6})(\sqrt{3}+1)} = \frac{6+2\sqrt{3}}{2(\sqrt{6})(\sqrt{3}+1)}$$

$$\text{CosA} = \frac{2\sqrt{3}(\sqrt{3}+1)}{2(\sqrt{6})(\sqrt{3}+1)} = \frac{1}{\sqrt{2}}$$

A = 45^0

Sol.10. (c)
angle opposite to largest side is always largest in a triangle.

$$CosC = \frac{a^2+b^2-c^2}{2ab}$$

$$CosC = \frac{6^2+10^2-14^2}{2.6.14}$$

$$CosC = \frac{-1}{2} \Rightarrow C = 120^0$$

Sol.11. (d)
if angles are in AP than 2B = A + C
and for triangle A + B + C = 180^0
by above two $\angle$B = 60^0
now cosine formula

$$CosB = \frac{a^2+c^2-b^2}{2ac}$$

$$\frac{1}{2} = \frac{a^2+c^2-b^2}{2ac}$$

$a^2 + c^2 - b^2 = ac$
$b^2 = a^2 + c^2 - ac$

Sol.12. (d)
we know that $18^2 + 24^2 = 30^2$
so triangle is right angled.
and Sin90^0 = 1

Sol.13. (a)
A = 30° , B = 45° and c = $\sqrt{3}+1$
then Angle C = 105°
noe by sine formula

$$\frac{SinA}{a} = \frac{SinC}{c} \Rightarrow \frac{Sin30^0}{a} = \frac{Sin105^0}{\sqrt{3}+1}$$

$$\Rightarrow \frac{1/2}{a} = \frac{\left(\dfrac{\sqrt{3}+1}{2\sqrt{2}}\right)}{\sqrt{3}+1} \Rightarrow a = \sqrt{2}$$

$$Area = \frac{1}{2}\times acSinB = \frac{1}{2}\times\sqrt{2}(\sqrt{3}+1)\frac{1}{\sqrt{2}}$$

$$Area = \frac{(\sqrt{3}+1)}{2}$$

Sol.14. (c)
I. $\dfrac{SinA}{a} = \dfrac{SinB}{b} = \dfrac{SinC}{c} = k$
Sin A + Sin B = Sin C
ak + bk = ck
a + b = c
it is not possible for a triangle.
II. A : B : C = 1 : 2 : 3
then A = 30° , B = 60° , C = 90°
for these angles
a : b : c = $\dfrac{1}{2}:\dfrac{\sqrt{3}}{2}:1 = 1:\sqrt{3}:2$

Sol.15. (a)
by sine rule

$$\frac{SinA}{a} = \frac{SinC}{c} \Rightarrow \frac{Sin45^0}{2\sqrt{2}} = \frac{SinC}{2}$$

C = 30°

Sol.16. (c)
Sin A – cosB = cosC
sinA = cosB + cosC

$$2\sin\frac{A}{2}\cos\frac{A}{2} = 2\cos\frac{B+C}{2}\cos\frac{B-C}{2}$$

$$2\sin\frac{A}{2}\cos\frac{A}{2} = 2\sin\frac{A}{2}\cos\frac{B-C}{2}$$

$$\cos\frac{A}{2} = \cos\frac{B-C}{2}$$

$$\frac{A}{2} = \frac{B-C}{2}$$

A = B – C
A + C = B
But A +B + C = 180
So B = 90

Sol.17. (a)

$$CosC = \frac{a^2+b^2-c^2}{2ab}$$

$$\Rightarrow Cos60^0 = \frac{(1+\sqrt{3})^2+(2)^2-c^2}{2(1+\sqrt{3}).2}$$

$$\Rightarrow \frac{1}{2} = \frac{(1+\sqrt{3})^2+(2)^2-c^2}{2(1+\sqrt{3}).2}$$

$$\Rightarrow \frac{1}{2} = \frac{(1+\sqrt{3})^2+(2)^2-c^2}{2(1+\sqrt{3}).2}$$

$$\Rightarrow 2+2\sqrt{3} = 4+2\sqrt{3}+4-c^2$$

$$\Rightarrow c = \sqrt{6}$$

$$CosA = \frac{b^2+c^2-a^2}{2bc}$$

$$CosA = \frac{4+6-(1+\sqrt{3})^2}{2.2.\sqrt{3}}$$

$$CosA = \frac{\sqrt{3}-1}{2\sqrt{2}} = Cos75^0$$

A = 75° , C = 60° then B = 45°

Sol.18. (b)

$$2aSin^2\left(\frac{C}{2}\right)+2cSin^2\left(\frac{A}{2}\right)=2a+2c-3b$$

⇒ a (1 – cos C) + c (1 – cos A) = 2a+2c−3b
⇒ (a + c) − (a cos C + c cos A) = 2a+2c−3b
⇒ a + c − b = 2a + 2c − 3b
⇒ a + c = 2b
Hence, a, b, c are in A.P.

Sol.19. (b)
Since, a, b, c are AP.
now by Sine formula
k Sin A , k Sin B , k Sin C
sin A, sinB, sin C are in AP.

Sol.20. (b)
given that p > 1
so take p = 2
for triangle a + b > c
take a = 4,b = 9, c = 10
half power of 4, 9, 10 are 2,3,3.2
so 2 + 3 – 3.2 is positive.

Sol.21. (b)
3tan(A+B)tanC=3tan120tan60 is negative
statement 1 is wrong.
tan(39+66) = tan105 that is negative
tan78 is positive
tan105 < tan78
statement 2 is correct
solve left hand side than we will get right hand side
so statement 3 is incorrect.

Sol.22. (c)
Cos A + Cos B + Cos C = $\sqrt{3}Sin\frac{\pi}{3}$

Cos A + Cos B + Cos C =3/2

we know that

$$\text{Cos A+Cos B+Cos C}=1+4\text{Sin}\left(\frac{A}{2}\right)\text{Sin}\left(\frac{B}{2}\right)\text{Sin}\left(\frac{C}{2}\right)$$

$$\Rightarrow \frac{3}{2} = 1+4Sin\left(\frac{A}{2}\right)Sin\left(\frac{B}{2}\right)Sin\left(\frac{C}{2}\right)$$

$$\Rightarrow Sin\left(\frac{A}{2}\right)Sin\left(\frac{B}{2}\right)Sin\left(\frac{C}{2}\right) = \frac{1}{8}$$

Sol.23. (d)

$$= Cos\left(\frac{A+B}{2}\right)Cos\left(\frac{B+C}{2}\right)Cos\left(\frac{C+A}{2}\right)$$

$$= Cos\left(90^0-\frac{C}{2}\right)Cos\left(90^0-\frac{A}{2}\right)Cos\left(90^0-\frac{B}{2}\right)$$

$$= Sin\left(\frac{C}{2}\right)Sin\left(\frac{B}{2}\right)Sin\left(\frac{A}{2}\right) = \frac{1}{8}$$

$$\Rightarrow Sin\left(\frac{A}{2}\right)Sin\left(\frac{B}{2}\right)Sin\left(\frac{C}{2}\right) = \frac{1}{8}$$

Sol.24. (b)
In triangle ABC, A + B + C = π

$$\Rightarrow \frac{A+B+C}{2} = \frac{\pi}{2}$$

$$\Rightarrow \frac{B}{2}+\frac{C}{2} = \frac{\pi}{2}-\frac{A}{2}$$

by taking sine both sides

$$\Rightarrow Sin\left(\frac{B+C}{2}\right) = Sin\left(\frac{\pi}{2}-\frac{A}{2}\right) = \cos\frac{A}{2}$$

also $\Rightarrow \tan\left(\frac{B+C}{2}\right) = \tan\left(\frac{\pi}{2}-\frac{A}{2}\right) = \cot\frac{A}{2}$

Sol.25. (b)

$$\cot\left(\frac{A}{2}\right)\cot\left(\frac{C}{2}\right)$$

$$\Rightarrow \sqrt{\frac{s(s-a)}{(s-b)(s-c)}\cdot\frac{s(s-c)}{(s-a)(s-b)}}$$

$$\Rightarrow \sqrt{\frac{s^2}{(s-b)^2}} = \frac{s}{s-b}$$

$$\frac{\frac{a+b+c}{2}}{\frac{a+b+c}{2}-b} = \frac{2b+b}{2b-b} = 3$$

[given that 2b = a + c]

Sol.26. (a)

$$\frac{Sin^2A+Sin^2B+Sin^2C}{Cos^2A+Cos^2B+Cos^2C} = 2$$

put A = 30° , B = 60° and C = 90°

these angles satisfies this equation.

so triangle is right angled.

Sol.27. (b)
sum of angles of a triangle = 180
(x – y) + x + (x + y) = 180
x = 60

Sol.28. (b)
by sine formula

$$\frac{SinA}{a} = \frac{SinB}{b}$$

$$\frac{2/3}{2} = \frac{SinB}{3}$$

B = 90°

Sol.29. (a)
angle made by a chord on centre is always double to made by a point on circumference.
cosβ = cos2α

$$\frac{1-\tan^2\alpha}{1+\tan^2\alpha}$$

Sol.30. (d)
Sin2A − Sin2B− Sin2C
= 2cos(A + B)sin(A−B) + sin(2A + 2B)
= 2cos(A + B)sin (A−B) +2sin (A+B) cos (A+B)
= 2cos(A + B)[sin(A − B) + sin(A + B)]
=−2cosC[2sinAsinB]
=−4sinAcosBcosC

Sol.31. (c)
angles are in the ratio 1 : 2 : 3
then angles are 30° , 60° , 90°
now Sin formula

$$\frac{SinA}{a} = \frac{SinB}{b} = \frac{SinC}{c}$$

$$\frac{1/2}{a} = \frac{\sqrt{3}/2}{b} = \frac{1}{c}$$

a : b : c = 1 : $\sqrt{3}$: 2 then

Sol.32. (d)

$$\tan A + \tan B = \frac{a}{b}+\frac{b}{a} = \frac{a^2+b^2}{ab} = \frac{c^2}{ab}$$

Sol.33. (d)
A, B, C are in AP
A + B + C = 180^0
2B = A + C
so B = 60^0

now Sin formula $\frac{SinB}{b} = \frac{SinC}{c}$

$$\frac{SinB}{SinC} = \frac{b}{c}$$

$$\frac{\sqrt{3}/2}{SinC} = \frac{\sqrt{3}}{\sqrt{2}}$$

C = 45^0 then Angle A = 75°

Sol.34. (d)
statement 1
b = c then ∠B = ∠C
A + B + C = 180°
2B + B + B = 180° ⇒ B = 45° = C then A = 90°
triangle is right angled.3
Statement 1 is incorrect.
Statement 2 A = 40° , B = 65° then C= 75°

by sine formula $\frac{\text{sinA}}{a} = \frac{\sin C}{c}$

$$\Rightarrow \frac{a}{c} = \frac{\sin A}{\sin C} = \frac{\sin 40°}{\sin 75°} = \sin 40° \cos ec 75°$$

Statement 2 is wrong.

Sol.35. (b)
statement 1
A + B + C = 180°
Sin(A + B) = Sin(180°−C)
Sin(90° + B) = Sin C
Cos B = Sin C
Sec B = Cosec C
Sin B = 1/3 (given)

then $Sec\ B = \frac{3}{2\sqrt{2}} = \text{cosec}\,C$

Statement 1 is wrong.
Statement 2
b cos B = c cos C

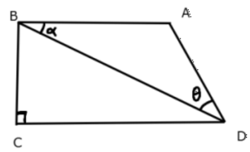
$$b = \frac{c\cos C}{\cos B}$$

by sine formula $\frac{\sin B}{b} = \frac{\sin C}{c}$

$$\frac{\sin B}{c\cos C}\cos B = \frac{\sin C}{c}$$

SinB cosB = sinC cosC
sin 2B = sin 2C
sin 2B−sin2C = 0
$2\cos(B+C)\sin(B-C) = 0$
cos(B+C) = 0 or Sin(B−C) = 0
B + C = 90° or B = C
triangle will be right angled or isosceles
But its already given that triangle is not right angled so triangle is isosceles
Statement 2 is correct.

Sol.36. (c)



Apply Sine formula in ΔABD

$$\frac{AD}{\sin\alpha} = \frac{AB}{\sin\theta} \Rightarrow AD\sin\theta = AB\sin\alpha$$

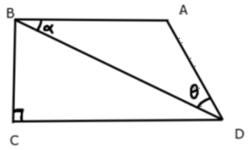
statement 1 is correct
Again apply Sine formula in ΔABD

$$\frac{BD}{\sin(180-(\alpha+\theta))} = \frac{AB}{\sin\theta} \Rightarrow \frac{BD}{\sin(\alpha+\theta)} = \frac{AB}{\sin\theta}$$

$$\Rightarrow BD\sin\theta = AB\sin(\alpha+\theta)$$

statement 2 is correct

Sol.37. (a)



given BC = p and CD = q

so BD = $\sqrt{p^2+q^2}$, cos α = $\dfrac{q}{\sqrt{p^2+q^2}}$,

sin α = $\dfrac{p}{\sqrt{p^2+q^2}}$

now apply Sine formula in ΔABD

$$\frac{BD}{\sin(\alpha+\theta)} = \frac{AB}{\sin\theta} \Rightarrow AB = \frac{\mathrm{BD}\sin\theta}{\sin(\alpha+\theta)}$$

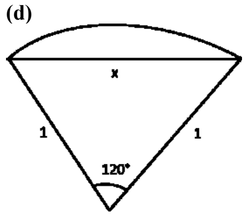
$$\Rightarrow AB = \frac{\sqrt{a^2+b^2}\,\sin\theta}{\sin\alpha\cos\theta - \cos\alpha\sin\theta}$$

put all above values

$$\Rightarrow AB = \frac{\sqrt{a^2+b^2}\,\sin\theta}{\frac{p}{\sqrt{p^2+q^2}}\cos\theta + \frac{q}{\sqrt{p^2+q^2}}\sin\theta}$$

$$\Rightarrow AB = \frac{(p^2+q^2)\sin\theta}{p\cos\theta + q\sin\theta}$$

Sol.38. (d)

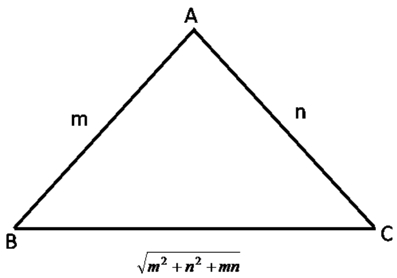


$$\cos\theta = \frac{a^2+b^2-c^2}{2ab}$$

$$\cos 120° = \frac{1+1-x^2}{2} \Rightarrow -\frac{1}{2} = \frac{1+1-x^2}{2}$$

$\Rightarrow x^2 = 3 \Rightarrow x = \sqrt{3}$ unit

Sol.39. (b)



$$\cos A = \frac{m^2+n^2-(m^2+n^2+mn)}{2mn}$$

$$\cos A = \frac{-mn}{2mn} = \frac{-1}{2}$$

A=120°

Sum of other two acute angles will be 60°.

Sol.40. (c)

Area of triangle = $\frac{1}{2}$ ac sinB

$= \frac{1}{2} \times 10 \times 4 \times \sin 30°$

$= \frac{1}{2} \times 10 \times 4 \times \frac{1}{2} = 10\ \text{cm}^2$

Sol.41. (d)

cos2A + cos2B + cos2C = – 1

2cos (A + B) cos (A – B) + cos 2C = – 1

2cos (π – C) cos (A – B) + 2 $\cos^2$C – 1 = – 1

–2cosC cos (A – B) + 2 $\cos^2$ C = 0

2cosC (cos C – cos (A – B) = 0

$$2\cos C\left(-2\sin\frac{C+A-B}{2}\sin\frac{C-A+B}{2}\right) = 0$$

$$-4\cos C\sin\left(\frac{\pi-2B}{2}\right)\sin\left(\frac{\pi-2A}{2}\right) = 0$$

–4cosC cosB cosA = 0

cosA cosB cosC = 0

Sol.42. (d)

sinA – cos B – cos C = 0

sin A = cosB + cos C

$2\sin\frac{A}{2}\cos\frac{A}{2} = 2\cos\frac{B+C}{2}\cos\left(\frac{B-C}{2}\right)$

∵ A + B + C = 180

B + C = 18 – A

$\frac{B+C}{2} = 90 - \frac{A}{2}$

$\operatorname{Cos}\left(\frac{B+C}{2}\right) = \cos\left(90 - \frac{A}{2}\right) = \sin\frac{A}{2}$

$2\sin\frac{A}{2}\cos\frac{A}{2} = 2\sin\frac{A}{2}\cos\frac{B-C}{2}$

$\cos\frac{A}{2} = \cos\frac{B-C}{2}$

$\frac{A}{2} = \frac{B-C}{2}$

$Á + C = B$

And A + B + C = 180°

So ∠ B = 90°

Sol.43. (d)

p = parimeter = 2

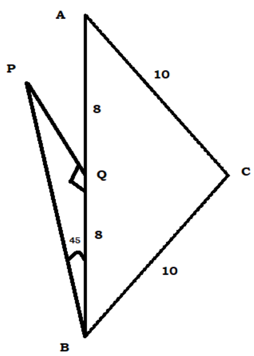
$2s\,(2s-2a)\,.\sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$

$4s\,(s\text{-}a).\sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$

$=4\sqrt{s(s-a)(s-b)(s-c)}$

= 4q

Sol.44. (c)



In ΔBPQ

$$\tan 45 = \frac{PQ}{8} \Rightarrow PQ = 8m$$

Sol.45. (b)

$$\cos C = \frac{100+100-256}{2.10.10}$$

$$\cos C = \frac{-56}{200} = -\frac{7}{25}$$

$$\sin C = \frac{24}{25}$$

$$\frac{AB}{\sin C} = \frac{16}{24/25} = \frac{50}{3}$$

Sol.46. (d)

triangle is isosceles

angle A = angle B

cosA + cosB + cosC

= 2cosA + cosC

$$\cos A = \frac{100+256-100}{2.10.16} = \frac{4}{5}$$

$$2\cos A + \cos C = \frac{8}{5} - \frac{7}{25} = \frac{33}{25}$$

What is cosA + cosB + cosC equal to?

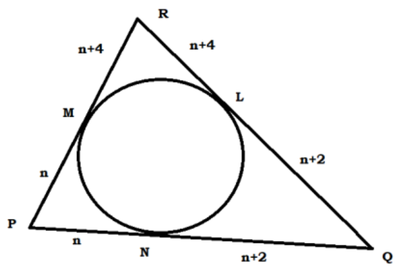
Sol.47. (a)

$$\cos C = \frac{a^2+b^2-c^2}{2ab} = \frac{16+9-4}{24} = \frac{7}{8}$$

$$\cos 3C = 4\cos^3 C - 3\cos C$$

$$= 4\left(\frac{343}{512}\right) - 3\left(\frac{7}{8}\right) = \frac{7}{128}$$

Sol.48. (c)



$$\cos P = \frac{PR^2 + PQ^2 + QR^2}{2.PR.PQ}$$

$$\frac{1}{3} = \frac{(2n+4)^2 + (2n+2)^2 + (2n+6)^2}{2(2n+4)(2n+2)}$$

By solving it we will get n = 8

Sol.49. (d)

If n=8 , then PR = 20, PQ = 18, QR = 22

Smallest side will be PQ = 18 unit

Sol.50. (c)

BC = a = √3

CA = b = 1

Perimeter = 6 x AM of sine of angles

$$a+b+c = 6 \times \frac{\text{SinA} + \text{SinB} + \text{SinC}}{3}$$

$$a+b+c = 2 \times (\text{SinA} + \text{SinB} + \text{SinC})$$

[we know that $\frac{\text{SinA}}{a} = \frac{\text{SinB}}{b} = \frac{\text{SinC}}{c} = k$

and SinA = ak , sinB = bk, sinC = ck]

$$a+b+c = 2 \times (ak + bk + ck)$$

$$a+b+c = 2 \times k(a+b+c)$$

$$1 = 2k$$

K = ½

$$\frac{\text{SinA}}{a} = \frac{\text{SinB}}{b} = \frac{\text{SinC}}{c} = \frac{1}{2}$$

$$\frac{\text{SinA}}{\sqrt{3}} = \frac{\text{SinB}}{1} = \frac{\text{SinC}}{c} = \frac{1}{2}$$

A = 60° and B = 30°, then C = 90°

$\frac{\text{Sin }90}{c} = \frac{1}{2}$, so c = 2

Perimeter = 2 + 1 + √3 = 3 + √3

Sol.51. (c)

By above solution C = 90° so triangle is right angle

And angles of triangle are in AP

Sol.52. (c)

$a^2 + b^2 + c^2 = ac + \sqrt{3}bc$

$a^2 + b^2 + c^2 - ac - \sqrt{3}bc = 0$

$a^2 - ac + \frac{c^2}{4} + b^2 - \sqrt{3}bc + \frac{3c^2}{4} = 0$

$\left(a - \frac{c}{2}\right)^2 + \left(b - \frac{\sqrt{3}}{2}c\right)^2 = 0$

$a - \frac{c}{2} = 0 \Rightarrow c = 2a$ ....(1)

$b - \frac{\sqrt{3}}{2}c = 0 \Rightarrow c = \frac{2b}{\sqrt{3}}$ ....(2)

By (1) and (2)

$2a = \frac{2b}{\sqrt{3}} = c$ or $\frac{a}{1/2} = \frac{b}{\sqrt{3}/2} = \frac{c}{1}$

By comparing $\frac{\text{SinA}}{a} = \frac{\text{SinB}}{b} = \frac{\text{SinC}}{c}$

sinA = ½ so A = 30

sin C = 1 so C = 90

and B = 60

so triangle is right angle triangle

Sol.53. (c)

by using above solution

$\frac{a}{1/2} = \frac{b}{\sqrt{3}/2} = \frac{c}{1}$

If c= 8 then a = 4 and b = 4√3

Area of right angle triangle = $\frac{1}{2}$ ×a×b

= $\frac{1}{2}$ ×4×4√3 = 8√3

Sol.54. (c)

The angles A, B and C of a triangle ABC are in the ratio 3:5:4.

Then A = 45 , B = 75 , C = 60

What is the value of a + b + √2c

$\frac{\text{Sin }45}{a} = \frac{\text{Sin }75}{b} = \frac{\text{Sin }60}{c} = \frac{1}{k}$

$\frac{1}{\sqrt{2}a} = \frac{\sqrt{3}+1}{2\sqrt{2}b} = \frac{\sqrt{3}}{2c} = \frac{1}{k}$

$a = \frac{k}{\sqrt{2}}, b = \frac{\sqrt{3}+1}{2\sqrt{2}}k, c = \frac{\sqrt{3}}{2}k$

$a + \sqrt{2}c = \frac{k}{\sqrt{2}} + \frac{\sqrt{3}}{\sqrt{2}}k = \left(\frac{\sqrt{3}+1}{\sqrt{2}}\right)k = 2b$

a + b + √2c = 3b

Sol.55. (a)

$a = \frac{k}{\sqrt{2}}, b = \frac{\sqrt{3}+1}{2\sqrt{2}}k, c = \frac{\sqrt{3}}{2}k$

$a^2 = \frac{k^2}{2}, b^2 = \frac{(\sqrt{3}+1)^2}{8}k^2, c^2 = \frac{3}{4}k^2$

$a^2 : b^2 : c^2$

$= \frac{k^2}{2} : \frac{(\sqrt{3}+1)^2}{8}k^2 : \frac{3}{4}k^2$

$= \frac{1}{2} : \frac{4+2\sqrt{3}}{8} : \frac{3}{4}$

$= 2 : 2+\sqrt{3} : 3$

Sol.56. (a)

In a triangle ABC, AB = 16 cm, BC = 63 cm and AC = 65 cm. What is the value of cos 2A + cos 2B + cos 2C?

we know that $16^2 + 63^2 = 65^2$

so triangle is right angled at angle B

B= 90 and A + C = 90

cos2A + cos2C + cos2B

$\Rightarrow 2\cos(A+C)\cos(A-C) + \cos 2B$

$\Rightarrow 2\cos 90 \cos(A-C) + \cos 180$

$\Rightarrow 0 - 1 = -1$

Sol.57. (a)

cotA·cotB·cotC > 0

i.e. all angles are acute

if any one angle become obtuse than cotangent of that angle will be negative.

so all angles are acute.

cotA·tanB·tanC > 0

i.e. all angles are acute

statement 1 is correct

if any one angle become obtuse than tangent of that angle will be negative.

so all angles are acute.

statement 2 is incorrect

Sol.58.

$$\frac{\cos A + \cos B}{\cos\left(\frac{A-B}{2}\right)} = \frac{2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)}{\cos\left(\frac{A-B}{2}\right)}$$

C = 60 so A + B = 120

$2\cos\left(\frac{120}{2}\right) = 1$

Sol.59. (a)

$\frac{a}{\cos A} = \frac{b}{\cos B} = \frac{c}{\cos C}$ ....(i)

but we know that

$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ .....(ii)

from (i) and (ii)

$\frac{a}{b} = \frac{\sin A}{\sin B} = \frac{\cos A}{\cos B}$

i.e. tanA = tanB

same as above tanB = tanC

if tanA = tanB = tanC

than A = B = C

triangle is equilateral

Area of equilateral triangle

$= \frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4}(6)^2 = 9\sqrt{3}$

Sol.60. (c)

roots of the equation $7x^2 - 6x + 1 = 0$ are tanα and tanβ

$\tan\alpha + \tan\beta = \frac{6}{7}$ and $\tan\alpha.\tan\beta = \frac{1}{7}$

$\tan(\alpha+\beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta} = \frac{\frac{6}{7}}{1-\frac{1}{7}} = 1$

so α + β = 45°

and 2α + 2β = 90°

in question it is given that 2α and 2β are the angles of triangle so triangle is right angled triangle.

from any statements or result we cannot conclude than α = β, so triangle is not isosceles.

Sol.61. (b)

we know that

$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

$\frac{\sin 75}{a} = \frac{\sin 45}{b} = \frac{\sin 60}{c}$

$\frac{\sqrt{3}+1}{2\sqrt{2}a} = \frac{1}{\sqrt{2}b} = \frac{\sqrt{3}}{2c}$

by above relation

$2a = \frac{(\sqrt{3}+1)}{\sqrt{3}}\sqrt{2}c$ and $b = \frac{\sqrt{2}}{\sqrt{3}}c$

$2a - b = \frac{(\sqrt{3}+1)}{\sqrt{3}}\sqrt{2}c - \frac{\sqrt{2}}{\sqrt{3}}c = \sqrt{2}c$

Sol.62. (b)

for a triangle we know that

A + B + C = 180°

A + B = 180° – C

tan(A + B) = – tanC

$\frac{\tan A + \tan B}{1 - \tan A \tan B} = -\tan C$

tanA + tanB = – tanC + tanA.tanB.tanC

tanA + tanB + tanC = tanA.tanB.tanC

given that tanA + tanB + tanC = k

so tanA.tanB.tanC = k

and cotA.cotB.cotC = 1/k

Sol.63. (c)

side CA is longest so its opposite angle B will be largesr

$\cos B = \frac{AB^2 + BC^2 - CA^2}{2AB.BC}$

$= \frac{9+25-49}{2.3.5} = -\frac{1}{2}$

$\cos B = -\frac{1}{2} \Rightarrow B = 120°$

So triangle is obtuse angled triangle and sum of other two angles is 60° i.e. acute.

Sol.64. (c)

By above solution

B = 120°

Sol.65. (a)

Area of triangle = $\frac{1}{2}$ × AB × BC × sinB

= $\frac{1}{2}$ × 3 × 5 × sin120° = 15√3/4 sq. cm

Sol.66. (b)

The angles A, B of a triangle ABC are in the ratio 3 : 1

Let angle A is 3x and B is x

$\frac{\text{Sin } A}{a} = \frac{\text{Sin } B}{b} \Rightarrow \frac{\sin 3x}{2k} = \frac{\sin x}{k}$

Sin3x = 2sinx

$3\sin x - 4\sin^3 x = 2\sin x$

$\sin x = 1/2$
$x = 30°$
angle $x = B = 30°$,

and $A = 3x = 90°$, $C = 60°$

Sol.67. **(c)**
by above solution
triangle is right angled triangle

angle $B = 30°$, $A = 90°$, $C = 60°$
so angles are in AP

Properties of Triangle 2011 2012 2013 Solution Video
Properties of Triangle 2014 2015 2016 Solution Video
Properties of Triangle 2017 2018 2019 Solution Video
Properties of Triangle 2020 2021 2022 Solution Video
Properties of Triangle 2023 2024 2025 Solution Video